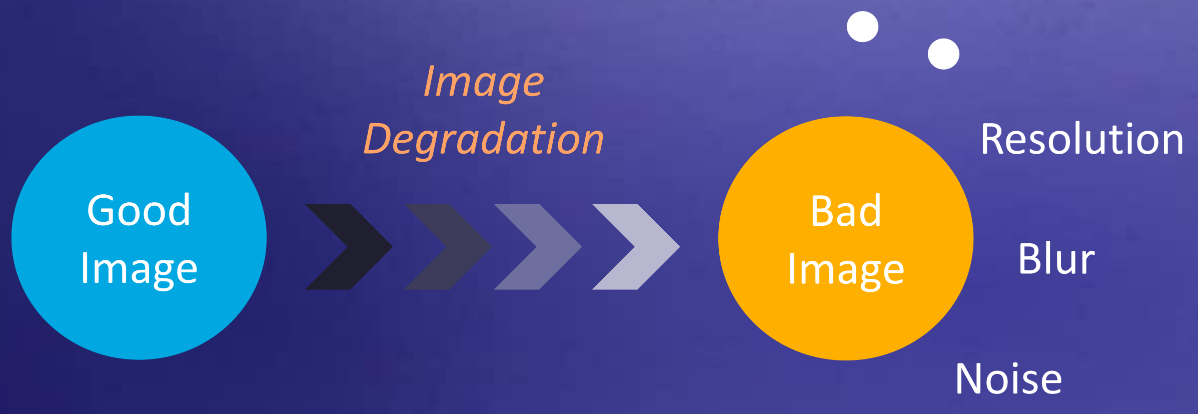
SemiCon AI Hackathon

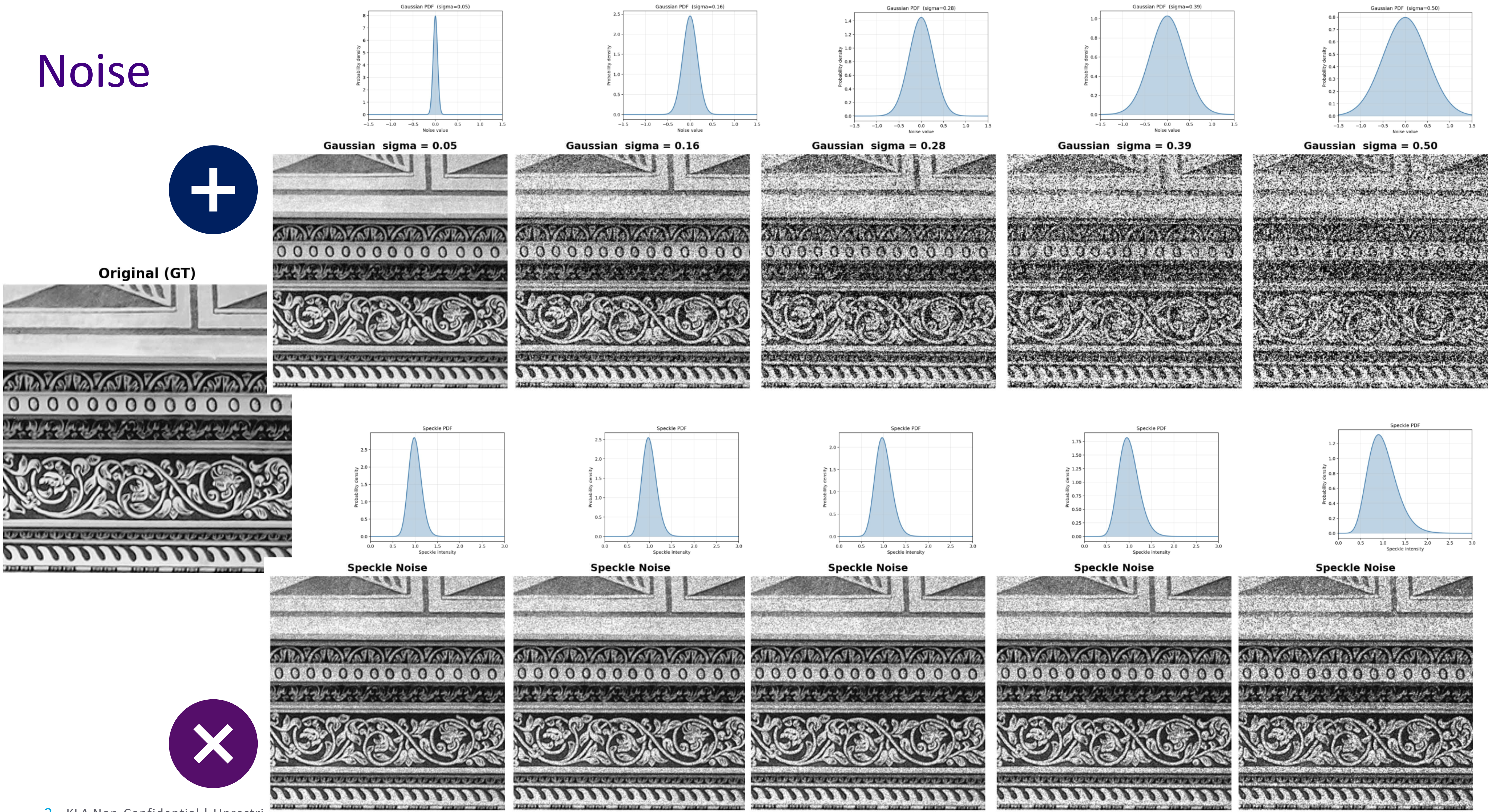
Image Restoration Task

30 July, 2026

What is a “Good image”?

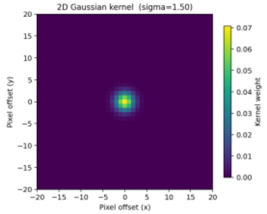


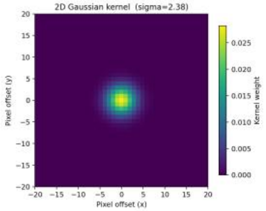
Noise

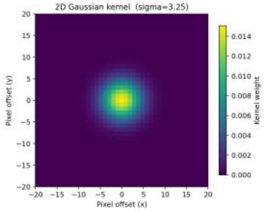


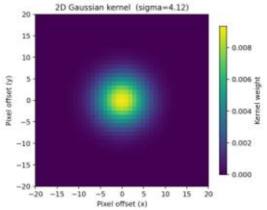
Gaussian PDF (sigma=0.05)
Gaussian PDF (sigma=0.16)
Gaussian PDF (sigma=0.28)
Gaussian PDF (sigma=0.39)
Gaussian PDF (sigma=0.50)
Probability density
Noise value
Gaussian sigma = 0.05
Gaussian sigma = 0.16
Gaussian sigma = 0.28
Gaussian sigma = 0.39
Gaussian sigma = 0.50
Original (GT)
Speckle PDF
Speckle intensity
Speckle Noise
Speckle Noise
Speckle Noise
Speckle Noise
Speckle Noise

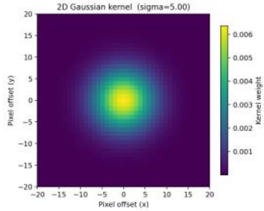
Blur











Blur sigma = 1.50

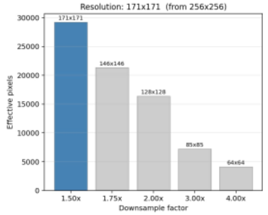
Blur sigma = 2.38

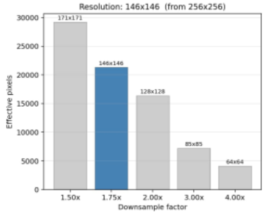
Blur sigma = 3.25

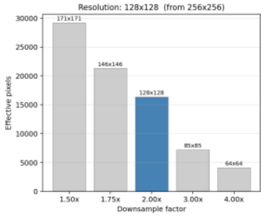
Blur sigma = 4.12

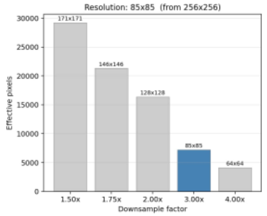
Blur sigma = 5.00

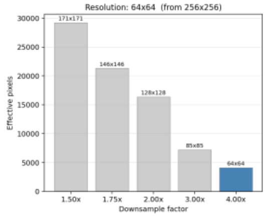
Original (GT)











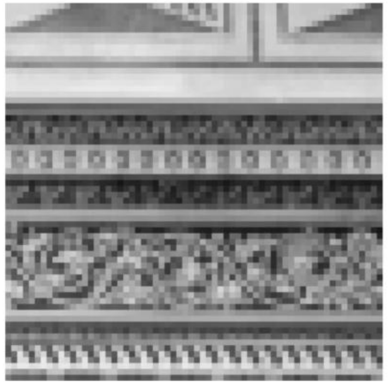
Downsample 1.50x

Downsample 1.75x

Downsample 2.00x

Downsample 3.00x

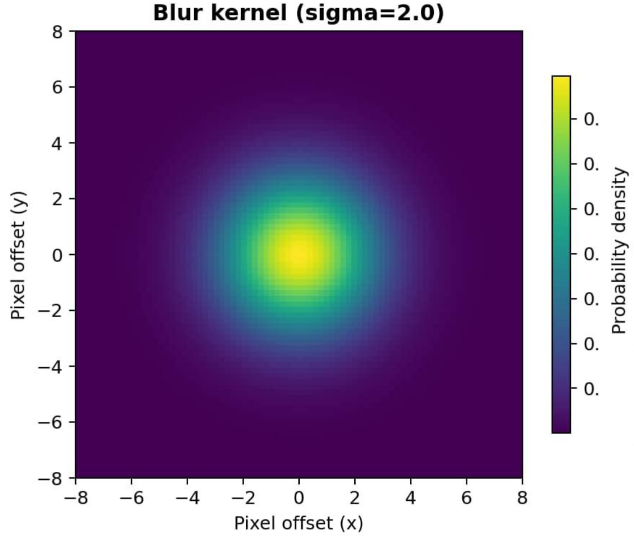
Downsample 4.00x

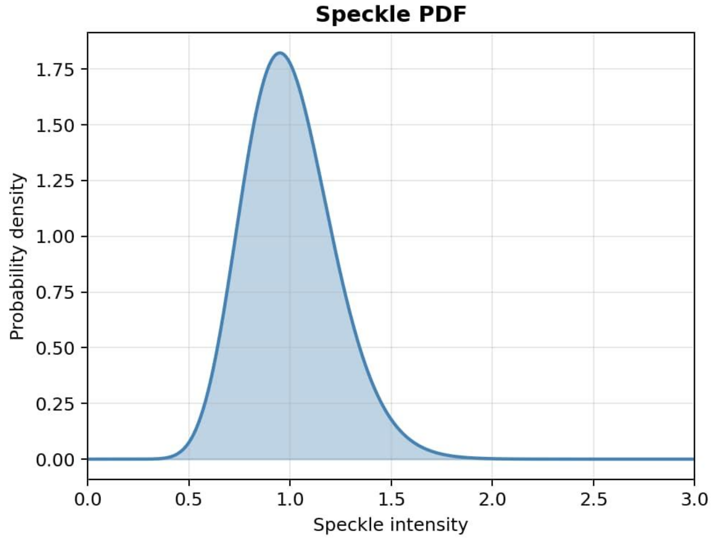


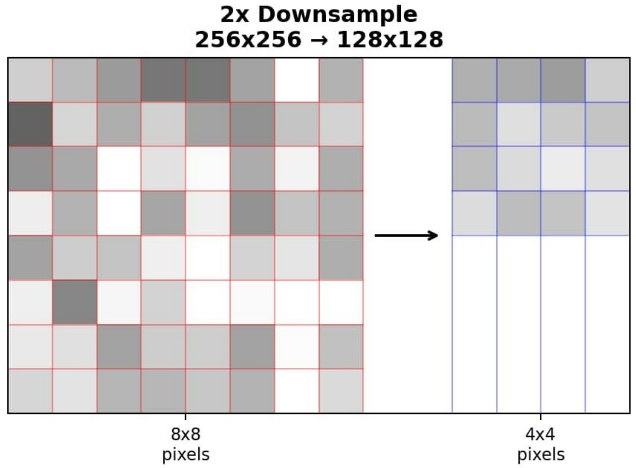
Resolution

Real world image degradation are more complex

Clean Image (No degradation)



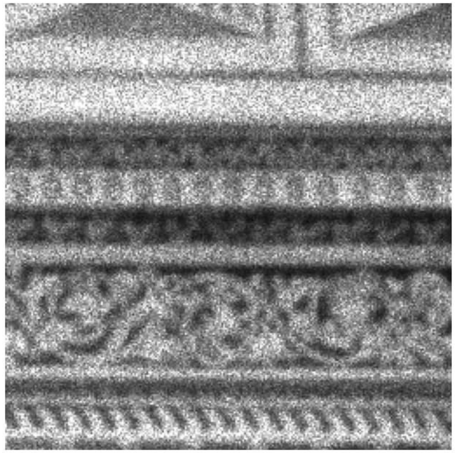




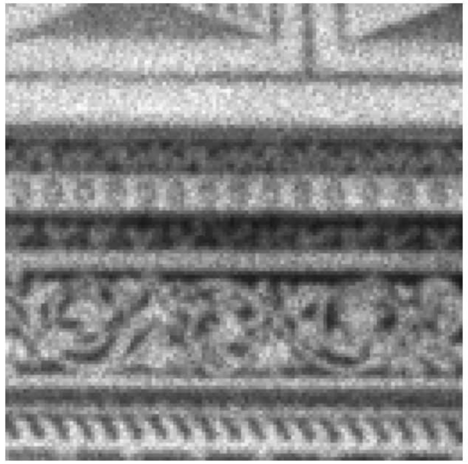
Original

After Blur

After Speckle



After Downsample

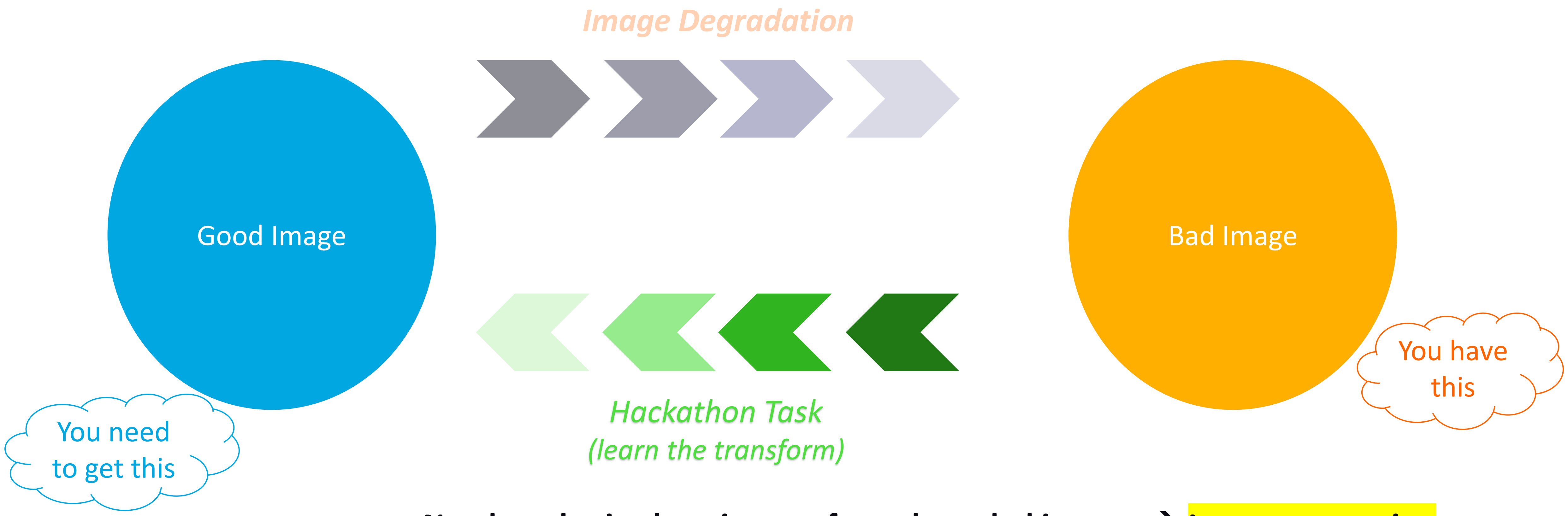


They may appear in any order!

What is the hackathon task?

Not a Trivial Problem ☺

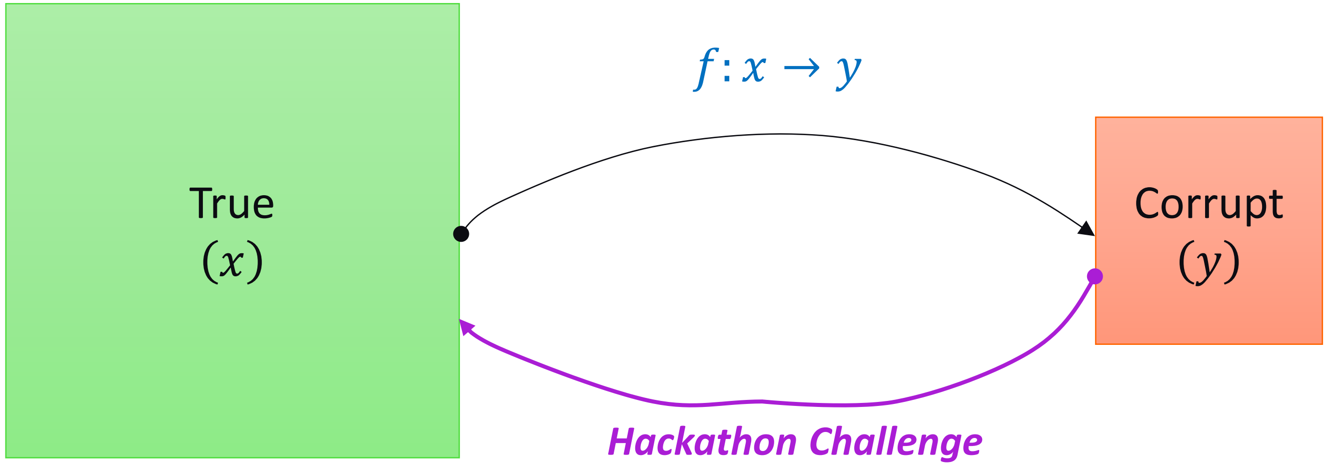
Speckle noise, down-sampling, additive gaussian noise
Do not read into the order of it



Need to obtain clean images from degraded images → Image restoration

Hackathon Problem

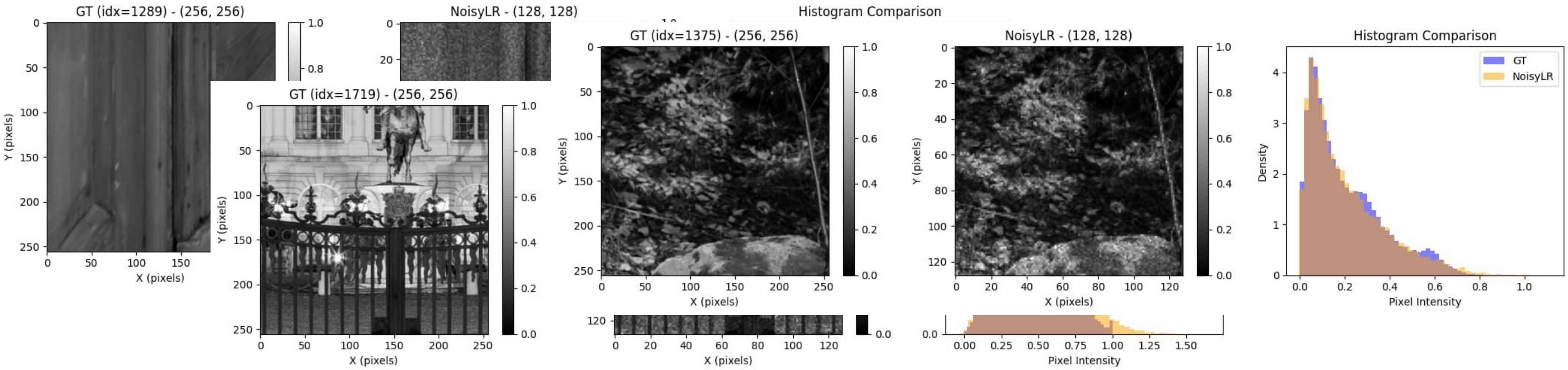
f: Speckle noise, down-sampling, additive gaussian noise



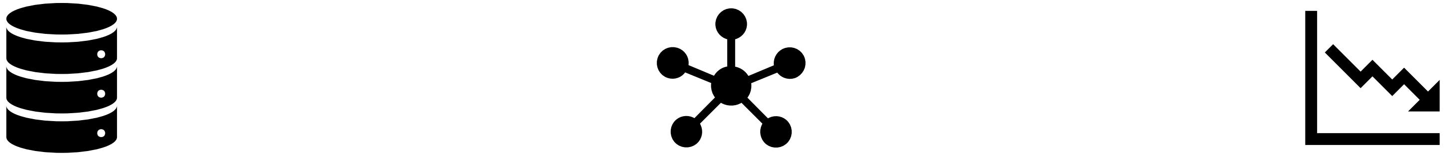
- Paired training data
- Test data (similar/dissimilar)

A few things to keep in mind:

- Image GL ranges: NoisyLR will have value beyond [0,1] whereas GT only has [0,1] – it is a feature not a bug!
- Images are normalized a-priori. Welcome to re-normalize. Note that GT is always [0,1].



While you build the model(s)…



Data[1] AI model[2] Loss[3]

Compute

Evaluations are not just on leaderboard statistics, but also on the emphasis put in each of these blocks.

Computation and Training hygiene is critical!

[1]T. Kumar, R. Brennan, A. Mileo and M. Bendechache, "Image Data Augmentation Approaches: A Comprehensive Survey and Future Directions," in IEEE Access, vol. 12, 2024.

[2]Zhai, Lujun, Yonghui Wang, Suxia Cui, and Yu Zhou. "A comprehensive review of deep learning-based real-world image restoration." *Ieee Access* 11 (2023): 21049-21067.

[3]Terven, J., Cordova-Esparza, DM., Romero-González, JA. *et al.* A comprehensive survey of loss functions and metrics in deep learning. *Artif Intell Rev* **58**, 195 (2025).

Thank You

Questions?

Image Restoration

(Broadly speaking) Restoration means recovering usable detail when images are corrupted with artifacts.

What are some image artifacts?

- Noise (Gaussian, Speckle,…), Resolution loss (down-sampling, blur,…)

Image restoration = recovering the “best estimate” of the original image from a degraded observation.

- Input: degraded image y
- Output: restored image $\hat{x}$ that approximates true image x

The core idea is the presence of a forward (read *’degradation’*) model. $f: x \to y$

But at observation you only have degraded images: y and the task (read ‘*restoration*’) for us is to estimate the true image x

This is a hard problem!

- Many possible x can explain the same y
- Noise/Down-sampling/Artifacts destroy information
- Real ‘*degradation*’ are often challenging to be modeled perfectly

How do we solve these?

If you have great priors → better chances of restoring images

Classical restoration methods use iterative methods and handcrafted priors to solve these problems (TV regularization, …)

AI learns these priors from examples, using millions of parameters.

- More expressive – learn priors we could not hand craft otherwise
- Direct image to image regression: Image in, Image out
- In advanced forms, it combines analytical iterative methods with learnt AI priors[1]!

[1] *V. Monga, et al., "Algorithm Unrolling: Interpretable, Efficient Deep Learning for Signal and Image Processing," in IEEE PM, vol. 38, no. 2, pp. 18-44, March 2021.*